

RISC-V VeeR EL2 Based Hardware Packet Buffering and Data Integrity SoC

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1. Problem Statement

Modern embedded and System-on-Chip (SoC) systems frequently receive data streams that must be temporarily stored and processed while maintaining data integrity. When incoming data arrives faster than the processor can process it, directly handling every incoming data element through software can increase processor workload and may result in data loss when the processing capacity is exceeded. In addition, corrupted data must be detected before it is processed by the system. Therefore, a hardware-assisted packet buffering and integrity mechanism is required to efficiently manage incoming data and provide reliable communication between the processor and peripheral hardware.

This project proposes the design and verification of a RISC-V VeeR EL2 based SoC with dedicated hardware IPs for packet buffering, packet monitoring, and data integrity checking. The SoC integrates the provided VeeR EL2 RISC-V processor with instruction memory, data memory, bus interconnect, UART, Timer, and GPIO as the mandatory system components. Three additional IP blocks—a Packet FIFO, Packet Counter, and CRC Checker—are integrated into the SoC as memory-mapped peripherals. The processor executes an application that controls the packet-handling subsystem, monitors packet status, and reports valid and corrupted packet information through UART.

2. Proposed SoC Architecture

The proposed SoC is organized around the VeeR EL2 RISC-V processor as the central processing unit. The VeeR EL2 is a 32-bit RV32IMC processor with a four-stage scalar in-order pipeline. The processor communicates with instruction memory, data memory, and peripheral IPs through the system interconnect. The VeeR EL2 core complex supports configurable system-bus interfaces, allowing the final interconnect implementation to be selected according to the provided processor RTL and project requirements.

Instruction Memory stores the application program executed by the processor, while Data Memory stores application data and temporary packet information. The bus interconnect performs address decoding and routes processor read and write transactions to the appropriate memory or peripheral based on the assigned memory map. UART is used to report packet status and system information, Timer provides timing functionality, and GPIO provides general-purpose input/output control.

The additional packet-processing IPs operate as follows:

- **Packet Buffering:** Incoming packet data is temporarily stored in the Packet FIFO so that data can be retained while the processor is processing previously received

information. FIFO status signals such as empty and full conditions are used to manage packet availability and overflow conditions.

- **Packet Monitoring:** The Packet Counter records packet-related activity such as accepted packets and, where required, error or overflow events. The processor can access the counter values through memory-mapped registers to monitor system activity.
- **Data Integrity Checking:** The CRC Checker calculates a cyclic redundancy check for received packet data and compares the calculated result with the corresponding received CRC value. A matching result indicates valid data, while a mismatch indicates a corrupted packet.

The overall data flow is:

Packet Input → Packet FIFO → CRC Checker → RISC-V Processor → UART

The Packet Counter operates alongside the packet-processing path to maintain packet statistics. The processor configures and monitors the additional IPs through their memory-mapped control and status registers. This architecture demonstrates hardware/software interaction, FIFO-based buffering, packet monitoring, data-integrity verification, memory-mapped peripheral integration, and SoC-level RTL design.

3. Selected IPs:

3.1 Packet FIFO

The Packet FIFO is a reusable buffering IP used to temporarily store incoming packet data. It provides controlled write and read operations and maintains status information such as full and empty conditions. An existing FIFO design developed previously will be adapted and integrated as a memory-accessible packet buffering component within the SoC.

3.2 Packet Counter

The Packet Counter is a lightweight hardware monitoring IP that maintains configurable counters for packet activity. It can record the number of received or accepted packets and can additionally track packet errors or FIFO overflow events. The counter values are made available to the RISC-V processor through memory-mapped registers.

3.3 CRC Checker

The CRC Checker is a hardware IP used to detect errors in received packet data. It calculates and compares the CRC value to identify valid and corrupted packets. The CRC result is accessible to the RISC-V processor through memory-mapped status registers.

The completed SoC will be verified through RTL simulation using normal packet transfers, multiple-packet buffering, CRC-valid packets, intentionally corrupted packets, and FIFO-full or overflow conditions. The processor application will read the packet-processing status and report the results through UART.